

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built production AI workflows using **Temporal.io**, **Prisma ORM**, **Node.js**, **React**, and **multi-model LLM pipelines** (OpenAI, Gemini, LangChain) across 7 deployments and 394 commits at an AI startup -- the same core stack Rollstack runs. Event-driven agentic platform at 2,000 async req/sec via Temporal + Kafka. RAG pipelines, LLM evals, and full-stack TypeScript from DB schema to UI. Ready to ship AI-powered data features at Rollstack.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Shipped **7 production deployments** across 10+ backend features using **TypeScript**, **Node.js**, **Express**, **Prisma ORM**, **PostgreSQL**, **Redis**, **Temporal.io** workflows, Pusher real-time pipelines, Auth0, Docker, **Sentry**, and Playwright. **394 commits**.
- Built **multi-model AI agent tooling across 3 providers and 8+ models** (Anthropic Claude, OpenAI GPT-4o, Google Gemini) via MCP; implemented document processing pipelines (PDF gen, email parsing via Microsoft Graph API, S3 file storage); reviewed PRs across the engineering team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation workflow (OCR, NLP, ML) improving accuracy 25%; automated AWS and GCP deployments with **Grafana + Sentry** observability and red-teaming support.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Hadoop + Spark + **Kafka** pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

AlgoPulse -- Agentic AI Platform 2025

Event-driven **LLM workflow orchestration** platform. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning directly to client. **Temporal.io** for durable workflow state + **Kafka** for fault-tolerant worker coordination -- **2,000 async inference req/sec** without state loss. TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

LLM Eval Harness 2025

Vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini on 2,000-example frozen test set. Tracked ExactMatch, grounded hallucination rate, p50/p95 latency, and cost/call with bootstrap significance testing -- analogous to **Langfuse**-style LLM observability. Streamlit dashboard for per-model failure analysis. Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Full-stack event-driven platform: MySQL outbox pattern, MongoDB TTL feed projections, Redis sorted sets for sub-ms leaderboard reads, Server-Sent Events for live updates. Multi-store consistency via **Testcontainers**; k6 benchmarks; **OpenTelemetry + Prometheus** observability; GitHub Actions CI/CD. TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

Languages: TypeScript, Python, Java, JavaScript, SQL, C

Stack: Node.js, React, NestJS, Next.js, Prisma ORM, Tailwind, Shadcn/UI, Express, Temporal.io, Kafka

AI/ML: OpenAI API, Gemini API, Claude API, LangChain, Langfuse, MCP, RAG, Hugging Face, PyTorch

Infra: Docker, Kubernetes, AWS, GCP, PostgreSQL, Redis, MongoDB, Sentry, Prometheus, GitHub Actions

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)